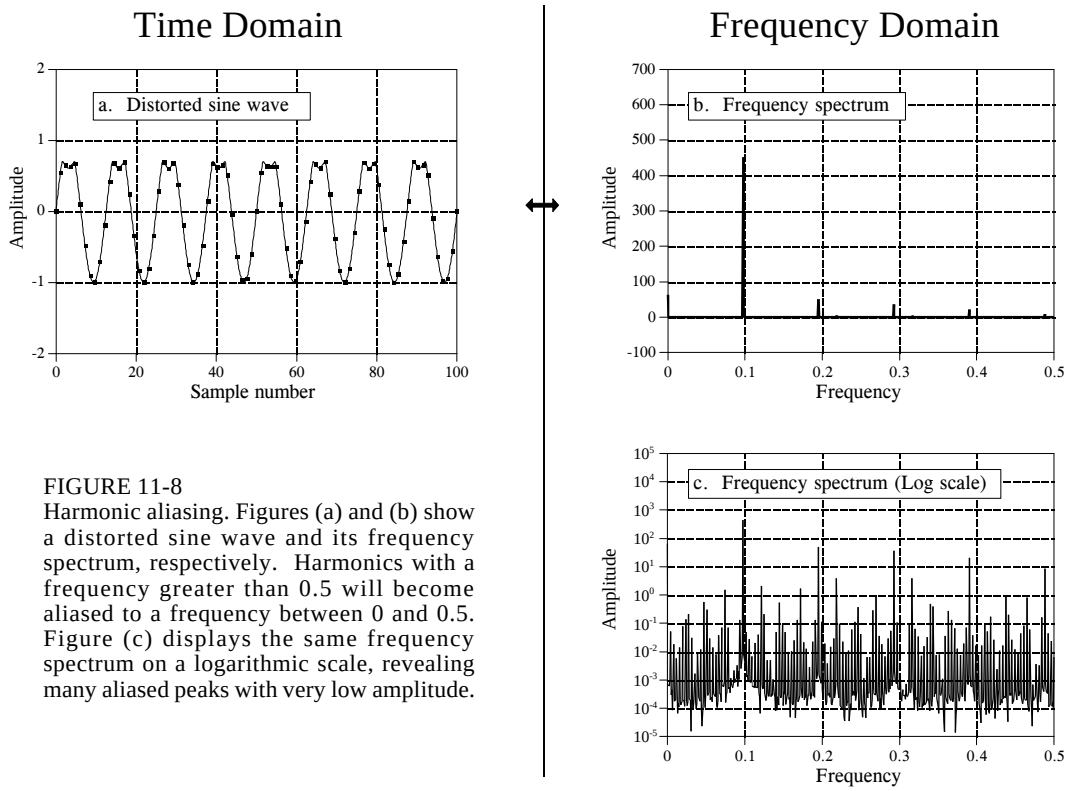

FIGURE 11-8
Harmonic aliasing. Figures (a) and (b) show a distorted sine wave and its frequency spectrum, respectively. Harmonics with a frequency greater than 0.5 will become aliased to a frequency between 0 and 0.5. Figure (c) displays the same frequency spectrum on a logarithmic scale, revealing many aliased peaks with very low amplitude.

example. Figure (a) is a pure sine wave, and (b) is its DFT, a single peak. In (c), the sine wave has been distorted by poking in the tops of the peaks. Figure (d) shows the result of this distortion in the frequency domain. Because the distorted signal is periodic with the same frequency as the original sine wave, the frequency domain is composed of the original peak plus harmonics. Harmonics can be of any amplitude; however, they usually become smaller as they increase in frequency. As with any signal, *sharp edges* result in *higher frequencies*. For example, consider a common TTL logic gate generating a 1 kHz square wave. The edges rise in a few nanoseconds, resulting in harmonics being generated to nearly 100 MHz, the *ten-thousandth* harmonic!

Figure (e) demonstrates a subtlety of harmonic analysis. If the signal is symmetrical around a horizontal axis, i.e., the top lobes are mirror images of the bottom lobes, all of the even harmonics will have a value of zero. As shown in (f), the only frequencies contained in the signal are the fundamental, the third harmonic, the fifth harmonic, etc.

All *continuous* periodic signals can be represented as a summation of harmonics, just as described. *Discrete* periodic signals have a problem that disrupts this simple relation. As you might have guessed, the problem is *aliasing*. Figure 11-8a shows a sine wave distorted in the same manner as before, by poking in the tops of the peaks. This waveform looks much less